

SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT

Fall 2023
Computer Organization & Assembly Language (CS-215T)
Assignment # 2

Semester: 3rd
Announced Date: 06-01-2024
Total Marks: 10
Instructor Name: Mr. Shakir Karim Buksh / Dr. Abdul Hameed Pitafi

Batch: 2022F
Due Date: 12-01-2024
Marks Obtained:

CLO #	Course Learning Outcomes (CLOs)	PLO Mapping	Bloom’s Taxonomy
CLO 2	Describe the design issues affecting the performance of a computer system.	PLO_2 (Knowledge for Computing Problem)	C2 (Understanding)

Q1. **[02 + 02 + 02 Marks]**

Let Speedup be denoted by “ S ”, fraction enhanced (or improved) be denoted by “ f_E ”, and factor of enhancement (or improvement) be denoted by “ K ” for a computer system.

- a. Consider a program have 40 percent of its code enhanced to run 2.3 times faster. **Estimate** the overall system speedup?
- b. Assume that a program have 60 percent of its code enhanced to get a system speedup of 5.3. **Predict** the factor by which the system is sped up.
- c. Suppose a program have a portion of its code enhanced to run 5 times faster to get a system speedup of 4.4. **Express** the fraction enhanced for the system.

Q2. **[04 Marks]**

Assume a computing module with the following components and MTTF (mean time to failure):

- 10 hard disks, each rated at 2,000,000 hour MTTF
- 1 disk controller with 500,000 hour MTTF
- 1 power supply with 200,000 hour MTTF
- 1 fan with 200,000 hour MTTF
- 1 cable for disk controller with 1,00,000 hour MTTF

Estimate the MTTF of the system as a whole, using the simplifying assumptions that the lifetimes are exponentially distributed and the failures are independent.